

# Introduction to Cyber Security

## Complex Engineering Problem (CEP)

*Part 1: Network Reconnaissance (Nmap) & Part 2: Traffic Analysis (Wireshark)*

**Submitted by: Aiman Salman**

**Roll No: 2025(S)-CYS105**

Course: Introduction to Cyber Security

Environment: VirtualBox 7.2 — Kali Linux (Attacker) and Windows 11 Host (Target)

## 1. Lab Environment

The reconnaissance exercise was carried out in a virtualized lab consisting of a Kali Linux attacker machine and the Windows 11 host machine acting as the target, connected via a VirtualBox Host-Only network. The table below lists the roles and IP addresses actually observed in this environment (these differ from the illustrative IPs in the assignment brief, since each student's virtual lab is addressed independently).

| Role             | System          | IP Address   | Network Adapter                   |
|------------------|-----------------|--------------|-----------------------------------|
| Attacker Machine | Kali Linux (VM) | 10.0.2.15    | NAT (eth0)                        |
| Target Machine   | Windows 11 Host | 192.168.56.1 | VirtualBox Host-Only (Ethernet 3) |

### 1.1 Verification of Attacker IP (Kali Linux)

The command 'ip a' was run on Kali Linux to confirm the attacker's own IP address, 10.0.2.15, on interface eth0.

```
(aiman@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:43:38:e1 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 86173sec preferred_lft 86173sec
    inet6 fd17:625c:f037:2:4c69:e3ce:10e5:3a64/64 scope global temporary dynamic
        valid_lft 86174sec preferred_lft 14174sec
    inet6 fd17:625c:f037:2:a00:27ff:fe43:38e1/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86174sec preferred_lft 14174sec
    inet6 fe80::a00:27ff:fe43:38e1/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Figure 1: Kali Linux (Attacker) — ip a showing IP 10.0.2.15

### 1.2 Verification of Target IP (Windows Host)

The command 'ipconfig' was run on the Windows host to confirm the target's IP address, 192.168.56.1, on the VirtualBox Host-Only adapter (Ethernet 3), which is the interface reachable from the Kali VM.

```
Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 3:

    Connection-specific DNS Suffix  . :
    Link-Local IPv6 Address . . . . . : fe80::a237:f955:9069:80de%16
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :
```

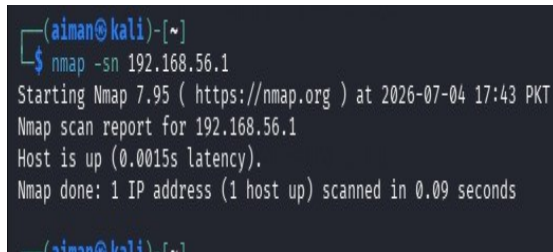
Figure 2: Windows Host (Target) — ipconfig showing IP 192.168.56.1

## 2. Part 1: Network Reconnaissance (Nmap)

### 2.1 Host Discovery

The following command was used from the attacker machine (Kali Linux, 10.0.2.15) to confirm the target host (192.168.56.1) was live and reachable:

```
nmap -sn 192.168.56.1
```



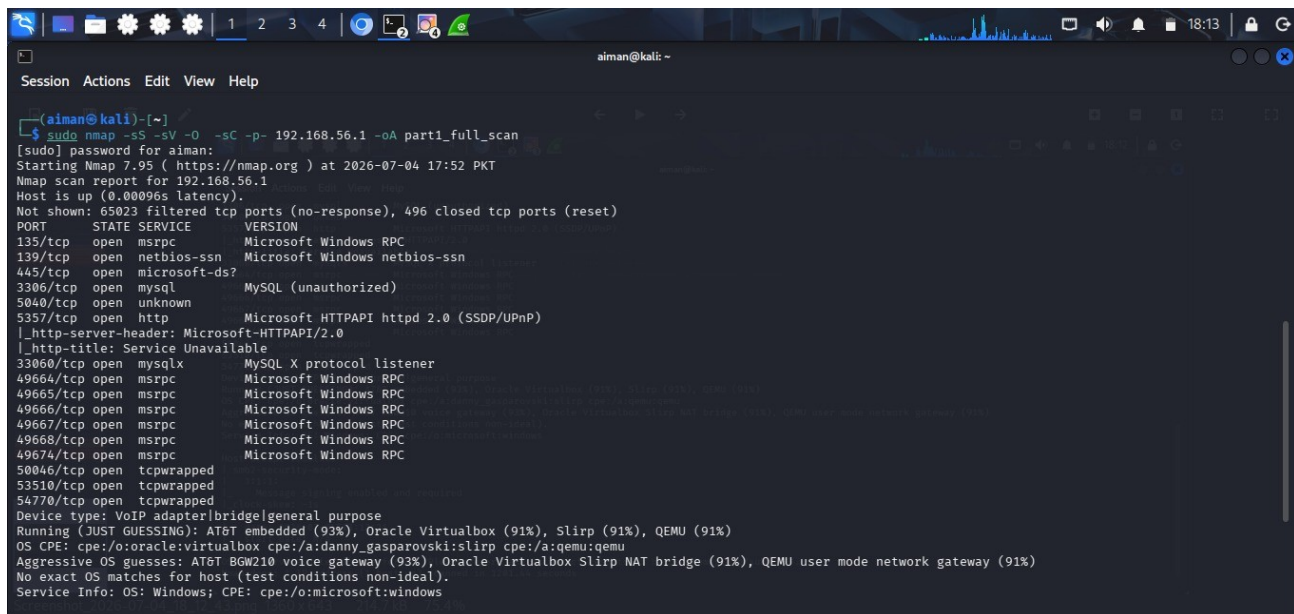
```
(aiman@kali)-[~]  
$ nmap -sn 192.168.56.1  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-04 17:43 PKT  
Nmap scan report for 192.168.56.1  
Host is up (0.0015s latency).  
Nmap done: 1 IP address (1 host up) scanned in 0.09 seconds  
(aiman@kali)-[~]
```

Figure 3: Host discovery scan confirming 192.168.56.1 is up

### 2.2 Comprehensive Port Scan, Service/Version Detection, OS Fingerprinting, and Script Scanning

A single comprehensive scan was run covering all 65,535 ports, service version detection, OS fingerprinting, and default NSE scripts, with output saved in all three required formats (Normal, XML, Grepable):

```
sudo nmap -sS -sV -O -sC -p- 192.168.56.1 -oA part1_full_scan
```



```
(aiman@kali)-[~]  
$ sudo nmap -sS -sV -O -sC -p- 192.168.56.1 -oA part1_full_scan  
[sudo] password for aiman:  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-04 17:52 PKT  
Nmap scan report for 192.168.56.1  
Host is up (0.00096s latency).  
Not shown: 65023 filtered tcp ports (no-response), 496 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
135/tcp   open  msrpc        Microsoft Windows RPC  
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn  
445/tcp   open  microsoft-ds?   
3306/tcp   open  mysql        MySQL (unauthorized)  
5040/tcp   open  unknown        
5357/tcp   open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)  
|_http-server-header: Microsoft-HTTPAPI/2.0  
|_http-title: Service Unavailable  
33060/tcp  open  mysqlx       MySQL X protocol listener  
49664/tcp  open  msrpc        Microsoft Windows RPC  
49665/tcp  open  msrpc        Microsoft Windows RPC  
49666/tcp  open  msrpc        Microsoft Windows RPC  
49667/tcp  open  msrpc        Microsoft Windows RPC  
49668/tcp  open  msrpc        Microsoft Windows RPC  
49674/tcp  open  msrpc        Microsoft Windows RPC  
50046/tcp  open  tcpwrapped     
53510/tcp  open  tcpwrapped     
54770/tcp  open  tcpwrapped     
Device type: VoIP adapter|bridge|general purpose  
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (91%), Slirp (91%), QEMU (91%)  
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny.gasparovski:slirp cpe:/a:qemu:qemu  
Aggressive OS guesses: AT&T BGW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (91%), QEMU user mode network gateway (91%)  
No exact OS matches for host (test conditions non-ideal).  
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows
```

Figure 4: Full scan results — open ports, services, and OS detection

### 2.3 Vulnerability Assessment

A dedicated vulnerability scan was performed using Nmap's built-in NSE vulnerability-scripting category:

```
sudo nmap --script vuln 192.168.56.1 -oA part1_vuln_scan
```

```

(aiman@kali)-[~]
$ sudo nmap --script vuln 192.168.56.1 -oA part1_vuln_scan
[sudo] password for aiman:
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-04 18:16 PKT
Nmap scan report for 192.168.56.1
Host is up (0.0042s latency).
Not shown: 995 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
3306/tcp   open  mysql
5357/tcp   open  wsddapi

Host script results:
|_smb-vuln-ms10-061: Could not negotiate a connection:SMB: Failed to receive bytes: ERROR
|_samba-vuln-cve-2012-1182: Could not negotiate a connection:SMB: Failed to receive bytes: ERROR
|_smb-vuln-ms10-054: false

Nmap done: 1 IP address (1 host up) scanned in 31.14 seconds

```

Figure 5: Vulnerability assessment scan output

## 2.4 Summary of Findings — Part 1

| Item                       | Result                                                                                                                                                                                          |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Live Host Discovered       | 192.168.56.1 (Windows Host) — Host is up                                                                                                                                                        |
| Attacker Machine           | 10.0.2.15 (Kali Linux)                                                                                                                                                                          |
| Open Ports                 | 135/tcp msrpc, 139/tcp netbios-ssn, 445/tcp microsoft-ds, 3306/tcp mysql, 5357/tcp wsddapi, 33060/tcp mysqlx, 49664-49674/tcp msrpc (dynamic RPC)                                               |
| Filtered/Closed Ports      | 995 ports reported as filtered (no response) during the vulnerability scan pass                                                                                                                 |
| Operating System Guess     | Windows (Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows)                                                                                                                              |
| Vulnerabilities Identified | smb-vuln-ms10-061 — connection could not be negotiated (inconclusive); samba-vuln-cve-2012-1182 — connection could not be negotiated (inconclusive); smb-vuln-ms10-054 — false (not vulnerable) |
| Output Files Generated     | part1_full_scan.nmap / .xml / .gnmap and part1_vuln_scan.nmap / .xml / .gnmap                                                                                                                   |

## 3. Part 2: Traffic Analysis (Wireshark)

### 3.1 Packet Capture Setup

Wireshark was run on the attacker machine (Kali Linux) on interface eth0 while the Nmap scans (host discovery, full comprehensive scan, and vulnerability script scan) were executed against the target (192.168.56.1). The full session was saved as wireshark\_eth0VKXVR3.pcapng. Four display filters were then applied to isolate the specific traffic patterns required by the assignment.

### 3.2 SYN Scan Packets

Filter used: `tcp.flags.syn==1` and `tcp.flags.ack==0`

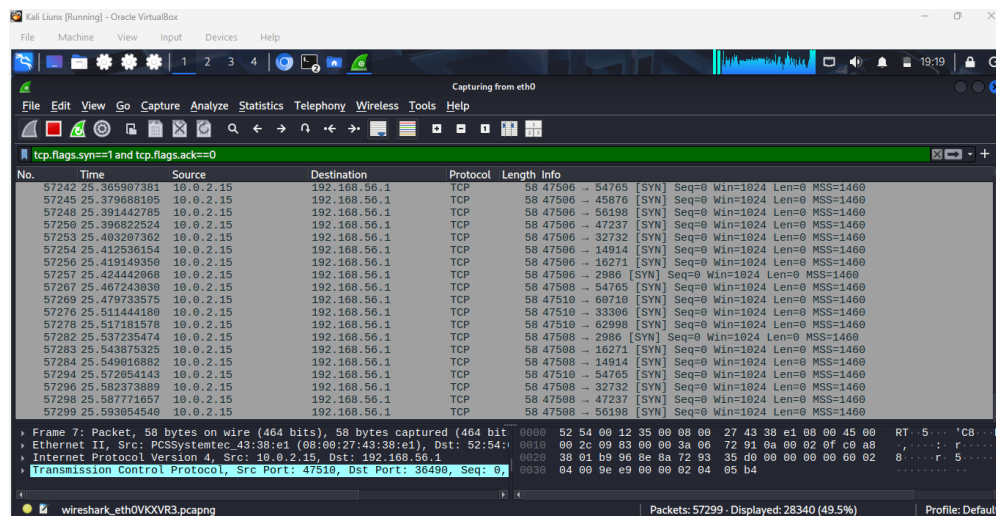


Figure 6: SYN scan packets - attacker (10.0.2.15) sending SYN probes with no payload (Len=0) to sequential/randomised destination ports on the target (192.168.56.1), Time approx 25.3s

Each frame (e.g., '47506 to 54765 [SYN] Seq=0 Win=1024 Len=0 MSS=1460') shows only the SYN flag set with a zero-length payload - the signature of Nmap's default SYN (stealth) scan, which never completes the TCP three-way handshake.

### 3.3 Operating System Fingerprinting Probes

Filter used: `tcp.flags==0x02` and `ip.dst==192.168.56.1`

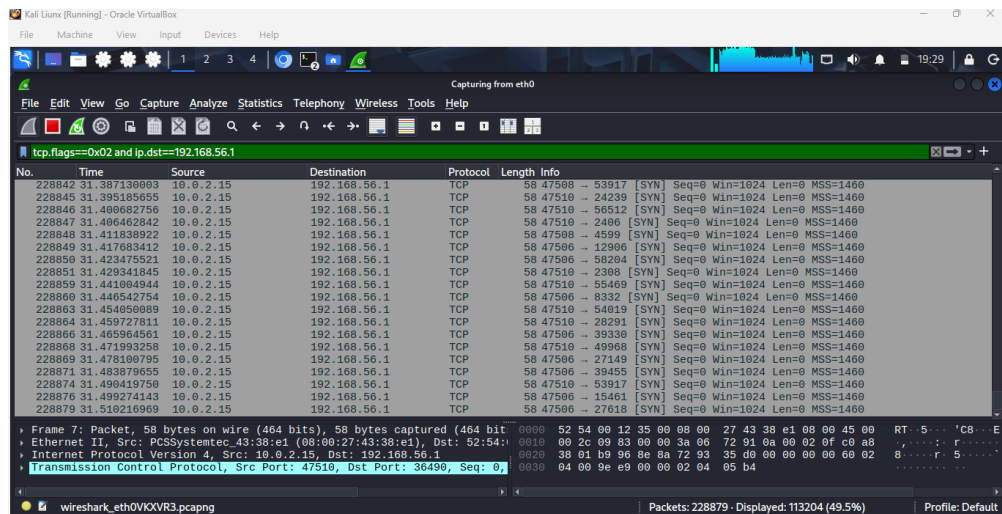


Figure 7: OS fingerprinting probes - raw SYN (flag 0x02) packets directed at the target, part of the -O OS-detection phase, Time approx 31.4s

Nmap's -O engine sends a series of specially-crafted TCP/ICMP probes (varying window sizes, TCP options, and flag combinations) to the target and analyses the responses to guess the operating system. The isolated 0x02-flag probes above correspond to this fingerprinting stage, occurring immediately after the initial SYN sweep.

### 3.4 Service / Version Detection Traffic

Filter used: `tcp.port==445`

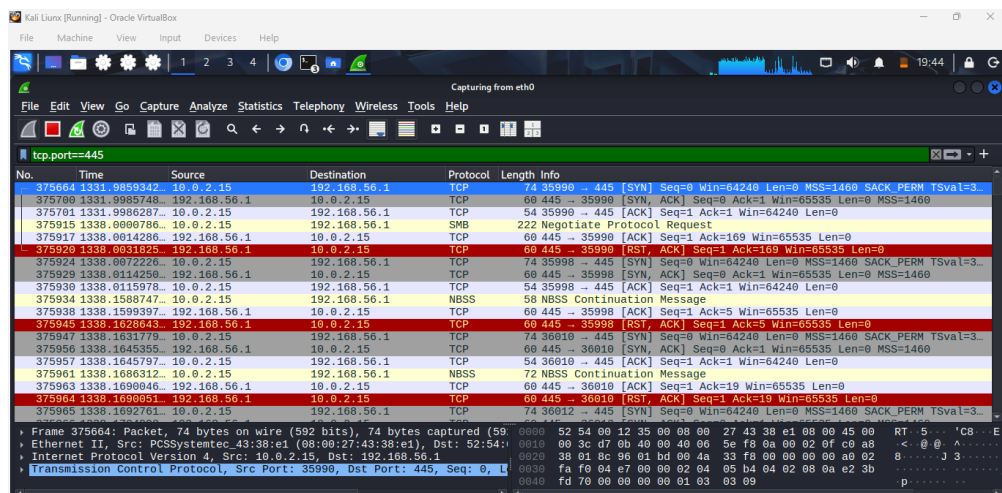


Figure 8: Service version detection - full TCP handshake followed by an SMB 'Negotiate Protocol Request/Response' exchange on port 445, Time approx 1331-1338s

Unlike the SYN scan, service/version detection (-sV) completes the full three-way handshake and exchanges application data. Here, port 445 (microsoft-ds/SMB) shows a completed handshake followed by an SMB Negotiate Protocol Request/Response, which Nmap uses to fingerprint the exact SMB service and version running on the target.

### 3.5 Application-Layer Script Probe Traffic

Filter used: `smb2`

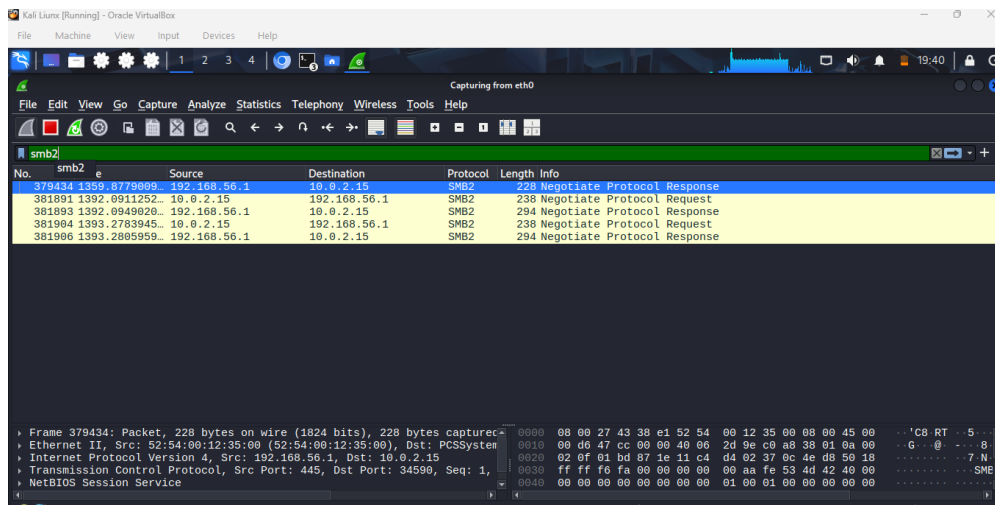


Figure 9: NSE script probe traffic - repeated SMB2 'Negotiate Protocol Request/Response' pairs generated by the vulnerability-scanning scripts (--script vuln), Time approx 1359-1394s

The repeated SMB2 Negotiate Protocol exchanges seen here were generated by Nmap's NSE scripts (-sC and --script vuln), which repeatedly negotiate the SMB dialect and query the service in order to test for known vulnerabilities (e.g., smb-vuln-ms10-061, smb-vuln-ms10-054). This application-layer conversation goes well beyond a simple port probe and is characteristic of script-engine traffic.

### 3.6 Volume and Distribution of Probed Traffic

Each filtered view above also reports the total packets captured in the session versus the packets matching that filter, which gives a sense of the scale and distribution of the reconnaissance traffic:

| Filter Applied                                          | Packets Displayed | Total Packets in Session |
|---------------------------------------------------------|-------------------|--------------------------|
| SYN scan (tcp.flags.syn==1 and tcp.flags.ack==0)        | 28,340            | 57,299                   |
| OS fingerprint probes (tcp.flags==0x02 and ip.dst==...) | 113,204           | 228,879                  |
| Service detection (tcp.port==445)                       | 94                | 382,304                  |
| Script probes (smb2)                                    | 5                 | 382,120                  |

This shows that the SYN sweep and OS-fingerprinting probes account for roughly half of all packets in the session (very high-volume, broad, and repetitive across almost every destination port), while service-version and script-probe traffic is comparatively tiny in packet count but far richer in application-layer content (full handshakes, SMB negotiation payloads) — reflecting Nmap's design of scanning broadly and cheaply first, then engaging deeply with only the ports found open.

Wireshark's Statistics → Protocol Hierarchy view corroborates this for the whole session (382,724 packets total):



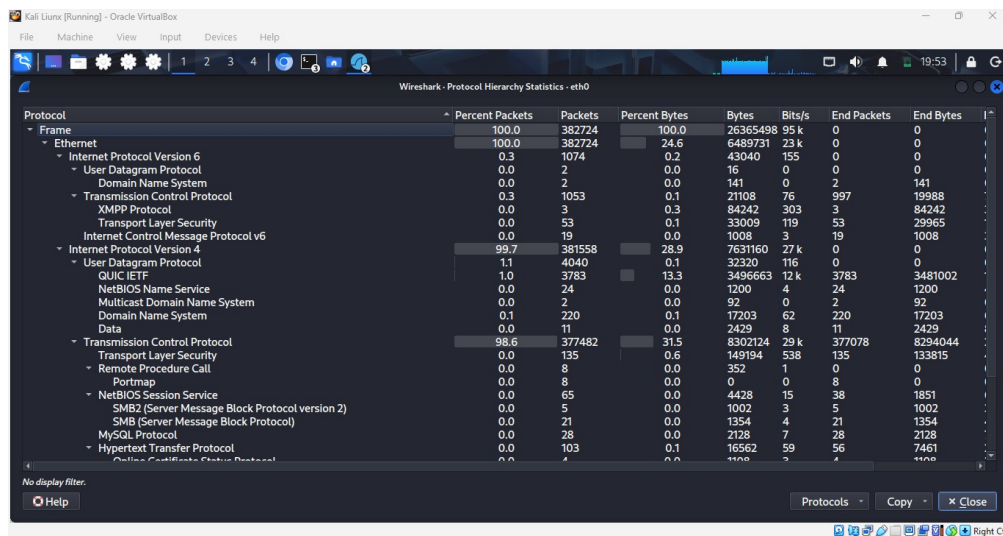


Figure 10: Protocol Hierarchy Statistics for the captured session (eth0) — TCP accounts for 98.6% of all packets (377,482), confirming the capture is dominated by the TCP-based Nmap scan traffic rather than background protocols

TCP makes up 98.6% of packets (377,482 of 382,724), consistent with the SYN/RST-ACK-heavy port-scanning traffic. Application-layer protocols invoked by Nmap's service/script detection — NetBIOS Name Service, NetBIOS Session Service, SMB, SMB2, Remote Procedure Call (Portmap), MySQL Protocol, and HTTP — each appear only in small numbers of packets, confirming they were generated by targeted probes against specific open ports rather than by continuous background traffic.

### 3.7 Temporal Sequence of Scan Phases

Comparing the timestamps of the four filtered views above shows the distinct, sequential phases Nmap executes internally during a single -sS -sV -O -sC scan, followed by the separate, later --script vuln scan:

| Approx. Time             | Phase                                                          | Evidence                                                                                            |
|--------------------------|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| t = 0-30s                | SYN port sweep (-sS)                                           | High-volume SYN-only packets across many destination ports (Figure 6)                               |
| t = 30-35s               | OS fingerprinting (-O)                                         | Raw flag-0x02 probes to the target immediately following the SYN sweep (Figure 7)                   |
| t = 1331-1338s (~22 min) | Service/version detection (-sV) during vulnerability scan pass | Completed handshakes with application-layer SMB Negotiate exchanges on ports 135/139/445 (Figure 8) |
| t = 1359-1396s (~23 min) | NSE script probes (-sC / --script vuln)                        | Repeated SMB2 Negotiate Protocol Request/Response cycles used for vulnerability checks (Figure 9)   |

The large gap between the initial SYN/OS-detection phase (first ~35 seconds) and the later service-detection/script phase (~22 minutes in) reflects the fact that the comprehensive full-port scan (part1\_full\_scan, which took 1201 seconds to complete) and the separate vulnerability scan (part1\_vuln\_scan) were captured in the same Wireshark session, with the vulnerability scan's negotiation and script traffic appearing well after the full scan's SYN sweep had finished.

### 3.8 Conclusion



The combined use of Nmap and Wireshark demonstrated the full reconnaissance lifecycle: discovering a live host, enumerating open ports and services, fingerprinting the operating system, checking for known vulnerabilities, and capturing/analysing each stage of the resulting network traffic - SYN sweep, OS-fingerprint probes, service/version detection, and NSE script probes - each isolated with its own display filter and placed in temporal sequence. The Windows target (192.168.56.1) was found to expose typical Windows networking services (RPC, SMB/NetBIOS, MySQL, WSD), with no confirmed exploitable vulnerabilities from the tested NSE scripts.